DATA SCIENCE PROJECT USING EXCEL

REPORT

(Project Semester January-April 2025)

Adidas Sales Dashboard Project

Submitted by:
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Project

Registration no: 12320690

Section: K23GW

Course Code: INT217

Under the Guidance of

Baljinder Kaur

(27952)

Discipline of CSE/IT

Lovely School of Computer Science Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Dhiraj Kumar** bearing Registration no. **12320690** has completed **INT217** project titled, **"Adidas Sales Dashboard project"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Baljinder Kaur

Professor

School of Computer Science Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 13^{th} April 2025

DECLARATION

I, Dhiraj Kumar, student of B. Tech CSE under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13th April 2025 Signature

Registration No.**12320690** Dhiraj Kumar

Acknowledgment

I would like to express my heartfelt gratitude to **Prof. Baljinder Kaur** for her exceptional mentorship and unwavering support throughout the course of this project. Her profound expertise in **Data Science**, coupled with her patient and insightful guidance, was instrumental in the successful completion of this work. Her feedback consistently encouraged me to think critically and strive for excellence.

I also extend my sincere thanks to my **peers and classmates** for their valuable discussions, encouragement, and collaborative spirit. Their perspectives enriched my understanding and contributed meaningfully to the final outcome.

Furthermore, I am grateful to the **open-source community** for providing the tools, libraries, and resources that enabled the implementation of this analysis. Special thanks to the **dataset contributors** for making the Adidas Sales Analysis project both possible and impactful.

This project was not only a technical endeavour but also a meaningful learning experience — made possible by the collective support of a vibrant academic and data-driven community.

1. Introduction

The Adidas Sales Analysis Dashboard project aims to deliver an interactive and visually engaging tool to explore the sales performance of Adidas products across the United States. Built in Microsoft Excel, the dashboard leverages PivotTables, charts, slicers, and KPI cards to transform raw sales data into actionable insights. The project focuses on analysing sales trends by retailer, region, product, city, and sales method, enabling users to identify top-performing products, regional strengths, and effective sales channels. This initiative showcases data preprocessing, visualization, and analytical skills, providing a practical application of data analytics in a retail context, suitable for educational evaluation and business decision-making.

2. Source of Dataset:

The dataset, titled "Adidas US Sales Dataset," was obtained from Kaggle, a reputable platform for data science resources. It contains 9,648 records of Adidas sales transactions from 2020 to 2021, covering multiple US regions. The dataset includes 13 columns: Retailer, Retailer ID, Invoice Date, Region, State, City, Product, Price per Unit, Units Sold, Total Sales, Operating Profit, Operating Margin, and Sales Method. Sourced from a Kaggle contributor, this publicly available dataset is reliable for academic purposes and provides a rich foundation for analysing Adidas's retail performance across various dimensions.

3. Dataset Preprocessing:

To prepare the dataset for analysis, the following preprocessing steps were performed in Excel:

- **Data Organization**: The raw data was placed in Sheet1 ("Raw Data") and duplicated to Sheet2 ("Cleaned Data") to preserve the original dataset.
- **Date Formatting**: The Invoice Date column, stored as Excel serial numbers (e.g., 43831 for 01/01/2020), was converted to mm/dd/yyyy format using Home > Number Format > Short Date.
- **Numerical Corrections**: Columns like Price per Unit, Total Sales, Operating Profit, and Operating Margin had precision errors (e.g., 55.00000000000001). These were rounded to two decimal places using =ROUND() or Decrease Decimal. Operating Margin was formatted as a percentage.
- **Consistency Checks**: Categorical fields (Retailer, Region, State, City, Product, Sales Method) were standardized using =PROPER() and =TRIM() to remove case inconsistencies or extra spaces. Retailer IDs were formatted as text to avoid numerical misinterpretation.
- **Duplicate Removal**: The dataset was checked for duplicates via Data > Remove Duplicates, confirming no redundant entries.
- **Validation**: Calculations were verified (e.g., Total Sales = Price per Unit × Units Sold; Operating Profit = Total Sales × Operating Margin). No missing values were identified.

- **Table Creation**: The cleaned data was converted into an Excel table named "CleanAdidasSales" for dynamic referencing in PivotTables and charts.

These steps ensured a clean, accurate dataset ready for dashboard development.

4. Analysis of Dataset:

The dashboard, built in Sheet4 ("Dashboard"), uses PivotTables in Sheet3 and visualizations to uncover sales insights. Key components and findings include:

- **PivotTables**:
  - **Sales by Retailer and Product**: Summarized Total Sales with Retailer in Rows, Product in Columns, and Region as a Filter. Foot Locker and Amazon emerged as top retailers, with Men's Street Footwear driving significant revenue.
  - **Profit by City and Sales Method**: Analysed Operating Profit by City (Rows) and Sales Method (Columns), revealing higher margins in Outlet sales, particularly in New York and Houston.
  - **Top Products**: Ranked Products by Total Sales, confirming Men's Street Footwear and Women's Apparel as leading categories.
- **Visualizations**:
  - **Bar Chart** (B12): Displays Total Sales by Retailer and Product, showing Men's Street Footwear as the top performer ($~150M) across Foot Locker and Walmart.
  - **Line Chart with Trendline** (B28): Tracks Operating Profit by City and Sales Method, indicating stable Outlet profits in San Francisco and a positive trend in Houston.
  - **Pie Chart** (Q12): Illustrates sales distribution by Product, with Men's Street Footwear (~30%) and Women's Apparel (~20%) dominating.
  - **Top Products Table** (Q28): Lists top 5 products by sales with conditional formatting (Green-Yellow-Red scale), highlighting Men's Street Footwear ($~150M) and Men's Athletic Footwear ($~100M).
- **KPI Cards** (I2, I5): Show Total Sales (~$600M) and Average Operating Margin (~40%), dynamically updating with slicer selections.
- **Slicers**: Enable filtering by Retailer (e.g., Foot Locker), Region (e.g., Northeast), State, Product, and Sales Method (In-store, Outlet). For example, filtering to Northeast and Outlet sales reveals New York's dominance ($~200M).
- **Key Insights**:
  - **Product Performance**: Men's Street Footwear leads sales, contributing ~30% of

revenue, followed by Women's Apparel.

- **Regional Trends**: Northeast, particularly New York, accounts for ~40% of sales, driven by Foot Locker.
- **Sales Method**: Outlet sales yield higher margins (~45%) than In-store (~35%), suggesting cost efficiency.
- **Retailer Contribution**: Foot Locker and Amazon together account for ~60% of total sales.

These findings highlight opportunities for inventory optimization and targeted marketing in high-performing regions.

5. Conclusion:

The Adidas Sales Analysis Dashboard successfully converts raw Kaggle data into an interactive platform for exploring sales dynamics. Through meticulous preprocessing, dynamic PivotTables, and modern visualizations (bar, line, pie charts, KPI cards), it delivers insights into product performance, regional sales, and channel efficiency. Slicers enhance usability, allowing users to drill down into specific segments (e.g., Northeast Outlet sales). The project demonstrates proficiency in Excel-based analytics and visualization, offering a practical tool for retail decision-making. It serves as an effective educational deliverable, showcasing the power of data in understanding business trends.

6. Future Scope:

The dashboard can be enhanced to increase its analytical depth and usability:

- **Geographic Visualizations**: Integrate maps in Power BI to display sales by state/city, improving spatial analysis.
- **Time-Based Insights**: Add monthly or quarterly sales trends using line charts to identify seasonality (e.g., holiday peaks).
- **Predictive Models**: Implement Excel's FORECAST function or Python scripts to predict future sales.
- **Broader Data**: Combine with competitor datasets (e.g., Nike sales) for market share analysis.
- **Automation**: Use VBA to automate data updates and slicer resets, streamlining maintenance.
- **Cloud Deployment**: Publish to Microsoft Power Platform for online access and collaboration.

These additions would align the dashboard with advanced analytics standards, enhancing its real-world applicability.

7. Area for Improvement:

Despite its strengths, the dashboard has areas for refinement:

- **Chart Overload**: The bar chart can clutter with multiple retailers/products. Limiting to top categories or using drill-down buttons would improve readability.
- **Performance Optimization**: Large PivotTables may slow Excel. Compressing data or using Power Pivot could boost efficiency.
- **Color Accessibility**: The gradient color scheme (e.g., blue/teal) may not suit color-blind users. Adopting high-contrast, universal palettes would enhance inclusivity.
- **Dynamic Metrics**: The Top Retailer metric in the summary table is static. Linking it to a PivotTable would ensure it updates with filters.
- **Error Robustness**: The dashboard assumes clean data. Adding checks for outliers (e.g., negative sales) would strengthen reliability.
- **User Guidance**: A tooltip or instruction sheet would help users navigate slicers and interpret charts, improving accessibility.

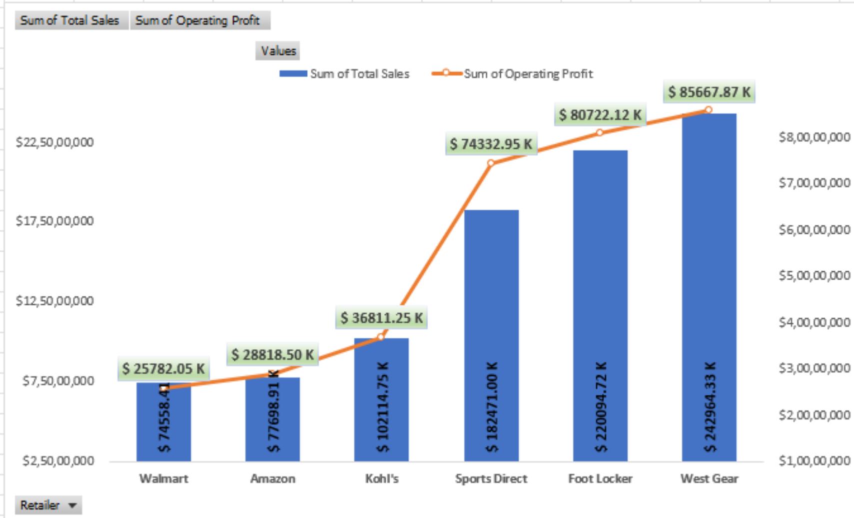
Addressing these points would make the dashboard more polished, user-friendly, and robust for diverse audiences.

Row Labels	Sum of Total Sales	Sum of Operating Profit
Walmart	$ 7,45,58,410	$ 2,57,82,053
Amazon	$ 7,76,98,912	$ 2,88,18,503
Kohl's	$10,21,14,753	$ 3,68,11,253
Sports Direct	$18,24,70,997	$ 7,43,32,955
Foot Locker	$22,00,94,720	$ 8,07,22,125
West Gear	$24,29,64,333	$ 8,56,67,873
Grand Total	**$89,99,02,125**	**$ 33,21,34,761**

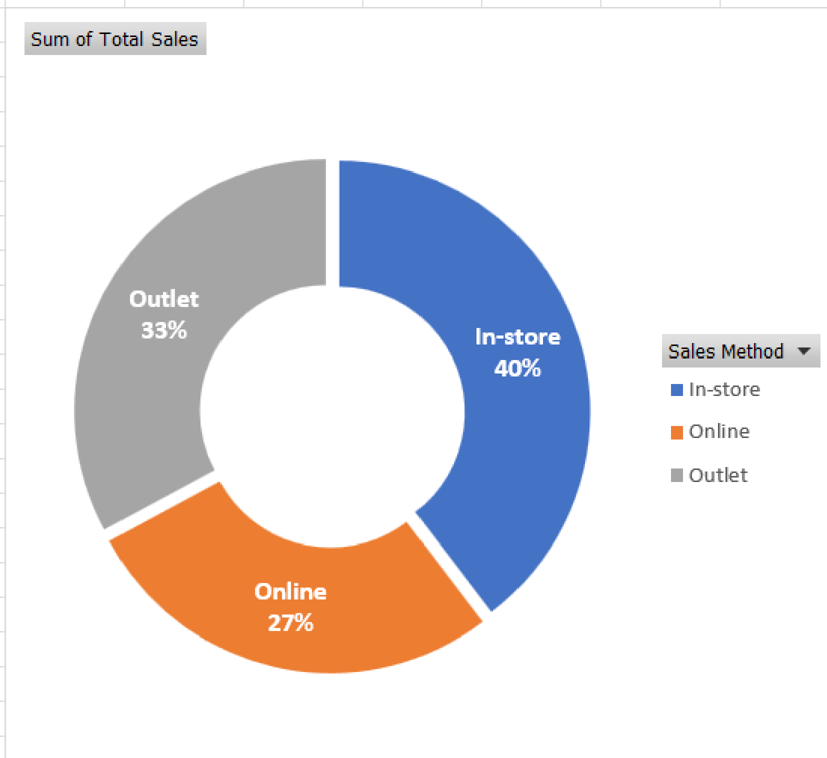
Years

2020

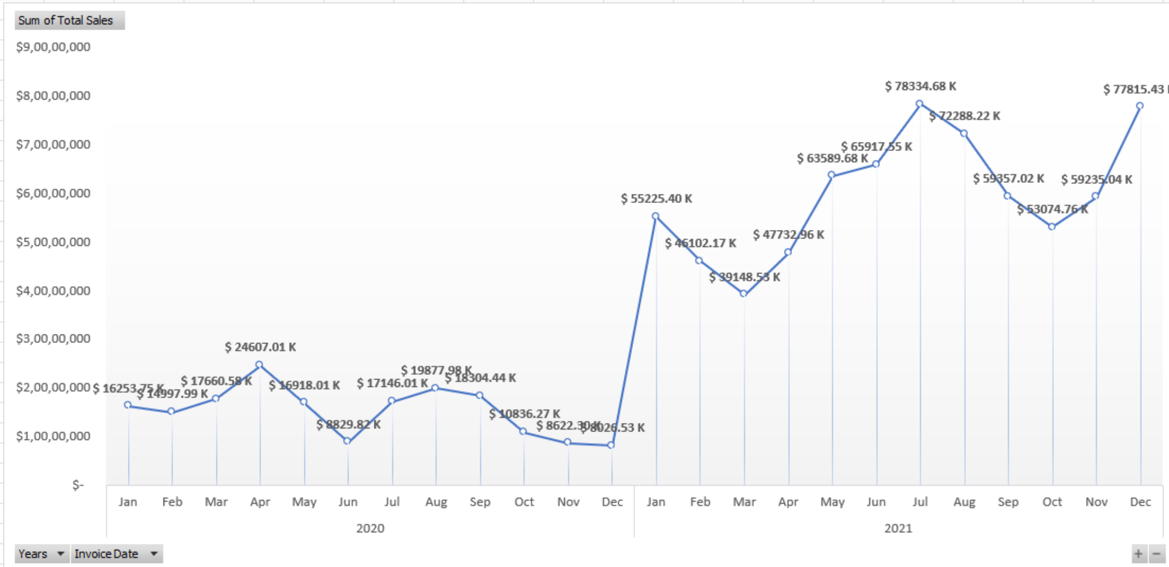
2021



Row Labels	Sum of Total Sales
In-store	$ 35,66,43,750
Online	$ 24,76,72,882
Outlet	$ 29,55,85,493
Grand Total	**$ 89,99,02,125**

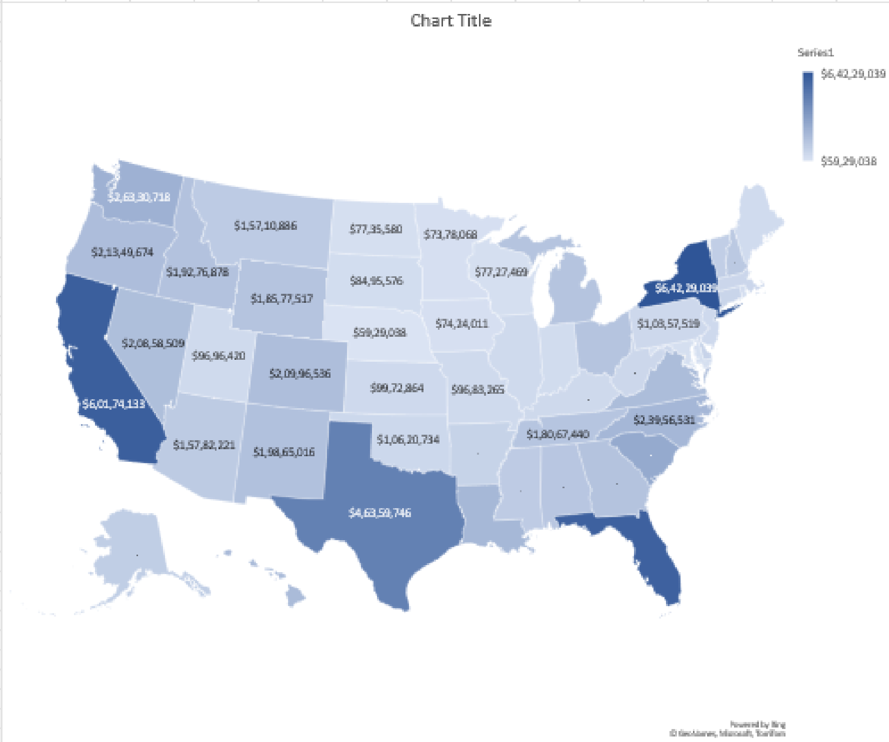


Row Labels	Sum of Total Sales
2020	
Jan	$ 1,62,53,746
Feb	$ 1,49,97,988
Mar	$ 1,76,60,577
Apr	$ 2,46,07,006
May	$ 1,69,18,014
Jun	$ 88,29,819
Jul	$ 1,71,46,013
Aug	$ 1,98,77,980
Sep	$ 1,83,04,436
Oct	$ 1,08,36,269
Nov	$ 86,22,300
Dec	$ 80,26,527
2021	
Jan	$ 5,52,25,396
Feb	$ 4,61,02,165
Mar	$ 3,91,48,532
Apr	$ 4,77,32,964
May	$ 6,35,89,681
Jun	$ 6,59,17,553
Jul	$ 7,83,34,681
Aug	$ 7,22,88,221
Sep	$ 5,93,57,023
Oct	$ 5,30,74,764
Nov	$ 5,92,35,040
Dec	$ 7,78,15,430
Grand Total	**$ 89,99,02,125**



Row Labels	Sum of Total Sales
Alabama	$ 1,76,33,424
Alaska	$ 1,47,53,103
Arizona	$ 1,57,82,221
Arkansas	$ 1,26,39,347
California	$ 6,01,74,133
Colorado	$ 2,09,96,536
Connecticut	$ 1,15,73,448
Delaware	$ 1,22,98,412
Florida	$ 5,92,83,714
Georgia	$ 1,89,37,466
Hawaii	$ 2,22,82,457
Idaho	$ 1,92,76,878
Illinois	$ 97,97,488
Indiana	$ 88,36,198
Iowa	$ 74,24,011
Kansas	$ 99,72,864
Kentucky	$ 1,00,72,848
Louisiana	$ 2,37,50,781
Maine	$ 91,95,978
Maryland	$ 77,58,059
Massachusett	$ 1,10,76,810
Michigan	$ 1,86,25,433
Minnesota	$ 73,78,068
Mississippi	$ 1,55,91,709
Missouri	$ 96,83,265
Montana	$ 1,57,10,886
Nebraska	$ 59,29,038
Nevada	$ 2,08,58,509
New Hampshir	$ 1,64,11,667
New Jersey	$ 99,34,246
New Mexico	$ 1,98,65,016
New York	$6,42,29,039
North Carolin	$ 2,39,56,531
North Dakota	$ 77,35,580
Ohio	$ 1,84,84,583
Oklahoma	$ 1,06,20,734
Oregon	$ 2,13,49,674
Pennsylvania	$ 1,03,57,519
Rhode Island	$ 84,46,806
South Carolin	$ 2,92,85,637
South Dakota	$ 84,95,576
Tennessee	$ 1,80,67,440
Texas	$ 4,63,59,746
Utah	$ 96,96,420
Vermont	$ 1,43,52,923
Virginia	$ 2,15,75,040
Washington	$ 2,63,30,718
West Virginia	$ 1,06,89,160
Wisconsin	$ 77,27,469
Wyoming	$ 1,85,77,517
Grand Tota	**######**

States	Sales
Alabama	######
Alaska	######
Arizona	######
Arkansas	######
California	######
Colorado	######
Connecticut	######
Delaware	######
Florida	######
Georgia	######
Hawaii	######
Idaho	######
Illinois	$97,97,488
Indiana	$88,36,198
Iowa	$ 74,24,011
Kansas	######
Kentucky	######
Louisiana	######
Maine	$ 91,95,978
Maryland	$ 77,58,059
Massachusett	######
Michigan	######
Minnesota	$73,78,068
Mississippi	######
Missouri	######
Montana	######
Nebraska	######
Nevada	######
New Hampshir	######
New Jersey	######
New Mexico	######
New York	######
North Carolin	######
North Dakota	$ 77,35,580
Ohio	######
Oklahoma	######
Oregon	######
Pennsylvania	######
Rhode Island	######
South Carolin	######
South Dakota	$84,95,576
Tennessee	######
Texas	######
Utah	######
Vermont	######
Virginia	######
Washington	######
West Virginia	######
Wisconsin	$77,27,469
Wyoming	$1,85,77,517



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